# Forecasting Realized Volatility with Graph Spillovers and Implied Volatility: A GNNHAR-IV Extension

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**Abstract**

This paper extends the graph neural network heterogeneous autoregressive framework of Zhang et al. (2025) by adding implied volatility (IV) as a forward-looking information channel. The original GNNHAR framework studies whether realized-volatility forecasts improve when the HAR model is augmented with graph-based spillover effects, multi-hop graph aggregation, nonlinear graph neural networks, and alternative estimation criteria such as MSE and QLIKE. We preserve that evaluation logic and construct HAR-IV, GHAR-IV, and GNNHAR-IV models for Dow 30 realized volatility forecasting. To distinguish genuine IV information from mere parameter expansion, we compare real-IV models with fake-IV controls that preserve the scale and dimension of IV features while breaking their time alignment. Motivated by preliminary simulation evidence on random adjacency matrices, we also compare GLASSO graph models with random-graph placebo models. This second placebo asks whether the estimated adjacency matrix itself contributes relation information, or whether graph terms mainly add transformed features and parameters. The IV source is AlphaQuery's stock-level implied-volatility data product, using the 30-day implied-volatility mean as the forward-looking option-market signal. The reproducible run covers 30 Dow constituents from 2021-02-26 to 2026-01-26, uses a chronological train-validation-test split, builds a GLASSO return network, and evaluates all forecasts using out-of-sample MSE, QLIKE, relative loss ratios, model confidence sets, Diebold-Mariano tests, and calm-versus-volatile regime splits. The best model by test QLIKE is GHAR-IV, with QLIKE 0.0007849, while the best model by test MSE is the validation-tuned GNNHAR3L-IV, with MSE 0.9591. The IV decomposition shows that real IV carries predictive information for linear HAR and GHAR specifications beyond a dimension-matched IV placebo. The random-graph tests give a more qualified interpretation: GHAR-IV with a random graph is nearly tied with GLASSO GHAR-IV, but GNNHAR3L-IV with a random graph deteriorates substantially. The evidence therefore supports the IV channel more directly than the linear graph channel, and it locates the clearest GLASSO-relation evidence in the nonlinear GNNHAR-IV specification.

**Keywords:** realized volatility; implied volatility; graph neural networks; HAR; GHAR; GNNHAR; QLIKE; model confidence set; volatility spillovers.

## 1 Introduction

Forecasting equity volatility is central to risk management, derivatives valuation, portfolio allocation, and stress testing. The heterogeneous autoregressive model of Corsi (2009) remains a strong benchmark because it expresses persistent realized volatility dynamics through daily, weekly, and monthly lag components. A limitation of the classical HAR specification is that it is largely asset-by-asset: it captures own-volatility persistence but does not explicitly represent volatility spillovers across assets. Zhang et al. (2025) address this limitation by embedding HAR features in a graph-based framework. Their study starts from graph HAR (GHAR), where neighbors are defined through a financial network, and develops GNNHAR models that use graph neural networks to capture nonlinear and multi-hop spillover effects. They evaluate forecasts with MSE and QLIKE, model confidence sets, pairwise forecast loss tests, and regime-specific comparisons.

This paper asks a complementary question. If graph spillovers improve the way past realized volatility is organized across assets, can the same framework benefit from a forward-looking market signal? We study this question by adding implied volatility (IV) to the GNNHAR framework. IV, extracted from option prices, reflects the market's forward-looking volatility assessment and is therefore conceptually different from realized volatility, which is backward looking. The resulting extension, denoted GNNHAR-IV, augments the original realized-volatility features with IV lag features at the same daily, weekly, and monthly HAR horizons. This design keeps the structure of Zhang et al.'s framework while adding a new economic information channel.

The key empirical concern is that IV and graph models contain more regressors and transformed features. Apparent gains could therefore arise from parameter expansion rather than genuine economic information. To address this, we introduce two placebo designs. Fake IV is obtained by permuting the IV feature tensor over dates, preserving the dimension and scale of the IV channel while destroying its timing relation with future realized volatility. Random graphs replace the GLASSO adjacency matrix with a density-matched random adjacency matrix. Comparing no-IV, real-IV, and fake-IV models decomposes the IV gain, while comparing GLASSO and random graphs tests whether graph performance reflects relation information rather than generic feature transformation.

Our contribution is fourfold. First, we provide a reproducible computational pipeline that implements HAR, GHAR, GNNHAR1L, GNNHAR2L, GNNHAR3L, and their IV counterparts on a Dow 30 panel. Second, we evaluate the IV extension using the same style of evidence emphasized by Zhang et al. (2025): MSE, QLIKE, relative loss ratios, model confidence sets (MCS), Diebold-Mariano (DM) tests, regime splits, and forecast distribution plots. Third, we show that IV improves linear HAR/GHAR specifications in a way that survives the fake-IV benchmark, and that validation-tuned GNNHAR-IV can outperform linear graph benchmarks by MSE while remaining close to GHAR-IV by QLIKE. Fourth, we include a random-relation placebo: random adjacency is not used because it is economically meaningful, but because it is a diagnostic for parameter and feature expansion.

The main finding is that GHAR-IV is the best-performing model by QLIKE in the tuned run, achieving a test QLIKE of 0.0007849 and a QLIKE ratio of 0.9549 relative to HAR. The validation-tuned GNNHAR3L-IV is the best-performing model by MSE, achieving test MSE 0.9591, below GHAR-IV's 1.0272 and GHAR's 1.0848. HAR-IV also improves on HAR, but GHAR-IV is better than HAR-IV, indicating that IV and graph spillovers are complementary. The fake-IV experiment strengthens the interpretation: for HAR and GHAR, real-IV gains are much larger than fake-IV gains. The random-graph experiment qualifies the graph interpretation.

In the linear GHAR-IV specification, random and GLASSO graphs are almost indistinguishable, so the linear graph-IV gain should not be over-interpreted as identified network structure. In contrast, GNNHAR3L-IV with the random graph performs much worse than GNNHAR3L-IV with GLASSO, indicating that the nonlinear graph extension is much more sensitive to the estimated relation matrix. Neural graph models therefore require careful validation tuning and placebo relation checks before they can be interpreted.

# 2 Relation to Zhang et al.'s GNNHAR Framework

Zhang et al. (2025) propose a customized GNN framework for forecasting multivariate realized volatilities with spillover effects. Their paper emphasizes three empirical questions: whether multi-hop neighbors add value, whether nonlinear spillovers improve forecasts, and whether QLIKE training can outperform MSE training because of heteroskedastic volatility errors. Their baseline comparison includes HAR, GHAR, and GNNHAR models with one, two, and three graph layers. Forecasts are evaluated out of sample by both MSE and QLIKE, and statistical evidence is summarized through loss ratios, MCS tests, DM-type comparisons, and market-regime analysis.

The present paper follows this architecture but modifies the information set. Let $v_{i,t}$ denote realized volatility of asset $i$ on day $t$, and let $x_{i,t}$ denote its implied volatility. The original GNNHAR framework uses lagged $v_{i,t}$ and graph-aggregated lagged realized volatility. We add lagged $x_{i,t}$ and graph-aggregated lagged IV. The economic motivation is straightforward: if option prices contain forward-looking information about expected volatility, then IV should help predict future RV even after conditioning on historical realized volatility and spillover effects. The econometric challenge is to distinguish that information from a larger feature set.

# 3 Model Specification

## 3.1 HAR and HAR-IV Features

For each asset $i$, the standard HAR feature vector contains daily, weekly, and monthly realized-volatility components:

$$H_{i,t-1}^{RV} = \left( v_{i,t-1},\ \frac{1}{5}\sum_{\ell=1}^{5} v_{i,t-\ell},\ \frac{1}{22}\sum_{\ell=1}^{22} v_{i,t-\ell} \right)'.$$

The IV extension mirrors this construction:

$$H_{i,t-1}^{IV} = \left( x_{i,t-1},\ \frac{1}{5}\sum_{\ell=1}^{5} x_{i,t-\ell},\ \frac{1}{22}\sum_{\ell=1}^{22} x_{i,t-\ell} \right)'.$$

The HAR and HAR-IV models are

$$\widehat{v}_{i,t}^{HAR} = \alpha_i + \beta_i' H_{i,t-1}^{RV},$$

and

$$\widehat{v}_{i,t}^{HAR\text{-}IV} = \alpha_i + \beta_i' H_{i,t-1}^{RV} + \delta_i' H_{i,t-1}^{IV}.$$

In the implementation, the linear models are estimated as pooled ridge regressions with standardized features, which stabilizes coefficient estimates in the panel setting.

### 3.2 GLASSO Graph Construction

Following the graph-spillover logic of Zhang et al. (2025), the network is estimated from training-window stock returns. Let $R_t$ be the vector of daily returns, $S$ the sample covariance matrix, and $\Theta$ the precision matrix. The graphical lasso solves

$$\widehat{\Theta} = \arg\min_{\Theta \succ 0} \left\{ \operatorname{tr}(S\Theta) - \log\det(\Theta) + \lambda \|\Theta\|_1 \right\}.$$

An undirected adjacency matrix $A$ is constructed by setting $A_{ij} = 1$ when $\widehat{\Theta}_{ij} \neq 0$ and $i \neq j$. The normalized graph operator is

$$W = D^{-1/2} A D^{-1/2},$$

where $D$ is the degree matrix. The empirical run selected GLASSO regularization $\lambda = 0.159968$ and produced edge density 0.442222.

### 3.3 GHAR-IV

GHAR augments HAR with graph-aggregated realized-volatility features:

$$\widehat{v}_{i,t}^{GHAR} = \alpha_i + \beta_i' H_{i,t-1}^{RV} + \gamma_i' \left( W H_{t-1}^{RV} \right)_i.$$

The IV extension includes both own IV and graph-aggregated IV:

$$\widehat{v}_{i,t}^{GHAR\text{-}IV} = \alpha_i + \beta_i' H_{i,t-1}^{RV} + \gamma_i' \left( W H_{t-1}^{RV} \right)_i + \delta_i' H_{i,t-1}^{IV} + \eta_i' \left( W H_{t-1}^{IV} \right)_i.$$

This specification is the most direct extension of Zhang et al.'s GHAR baseline because it retains graph spillovers while adding a forward-looking information channel.

### 3.4 GNNHAR-IV

The GNNHAR model replaces the linear graph aggregation in GHAR with nonlinear graph convolutional layers. Let $Z_t^{(0)}$ denote the input feature tensor. In the no-IV model, $Z_t^{(0)} = H_{t-1}^{RV}$; in the IV model, $Z_t^{(0)} = [H_{t-1}^{RV}, H_{t-1}^{IV}]$. A graph layer is represented as

$$Z_t^{(\ell+1)} = \sigma\left( W Z_t^{(\ell)} B^{(\ell)} + b^{(\ell)} \right),$$

where $B^{(\ell)}$ is a trainable weight matrix and $\sigma(\cdot)$ is a ReLU nonlinearity. The output layer maps node embeddings to positive volatility forecasts using a softplus link. One, two, and three graph layers correspond to GNNHAR1L, GNNHAR2L, and GNNHAR3L, respectively. This multi-layer design parallels Zhang et al.'s multi-hop setup. The empirical implementation therefore includes both no-IV neural graph models and their IV-augmented counterparts:

$$\text{GNNHAR}k\text{L-IV}: \quad Z_t^{(0)} = [H_{t-1}^{RV}, H_{t-1}^{IV}], \qquad k \in \{1,2,3\}.$$

These models are not linear GHAR-IV regressions. They are nonlinear graph neural networks whose first-layer inputs contain both realized-volatility HAR features and AlphaQuery IV HAR features, and whose graph propagation matrix is the same GLASSO network used in GHAR.

### 3.5 Forecast Losses and Estimation Criteria

Forecast performance is evaluated using MSE and QLIKE:

$$L_{MSE}(v,\widehat{v}) = (v-\widehat{v})^2,$$

$$L_{QLIKE}(v,\widehat{v}) = \frac{v}{\widehat{v}} - \log\left(\frac{v}{\widehat{v}}\right) - 1,$$

with forecasts clipped away from zero. The distinction between estimation criterion and forecast loss is important. Linear models are estimated by MSE, while the neural models include both MSE-trained and QLIKE-trained variants. All models are evaluated out of sample using both MSE and QLIKE.

### 3.6 Placebo Decompositions: Fake IV and Fake Relations

Let $L_0$, $L_{IV}$, and $L_{fake}$ denote the test loss of a no-IV model, its real-IV extension, and its fake-IV extension. We report

$$\Delta_{total} = L_0 - L_{IV}, \quad \Delta_{info} = L_{fake} - L_{IV}, \quad \Delta_{param} = L_0 - L_{fake}.$$

$\Delta_{total}$ is the total IV improvement, $\Delta_{info}$ is the genuine information gain, and $\Delta_{param}$ is the parameter expansion gain. A credible IV result should have $\Delta_{info} > 0$, not merely $\Delta_{total} > 0$.

The same logic applies to the graph relation matrix. In small samples, random adjacency matrices can occasionally perform as well as economically motivated or estimated adjacency matrices. That outcome is not a reason to prefer random graphs economically; it indicates that graph terms can behave as generic feature transformations. In GHAR, for example,

$$\mathbb{E}(v_{i,t} \mid \mathcal{F}_{t-1}) = \alpha_i + H_{i,t-1}^{RV}\beta + (WH_{t-1}^{RV})_i\gamma,$$

so even a fake $W$ creates extra regressors $(WH_{t-1}^{RV})_i$. We therefore compare GLASSO $W$ with a density-matched random $W^{rand}$. If the GLASSO model does not outperform the random-graph placebo, the result should be interpreted as parameter or feature expansion rather than identified spillover structure. If GLASSO materially outperforms the random graph, the evidence is stronger that the estimated relation matrix carries useful cross-sectional information.

Table 1: Data and full-run configuration

| Item | Value |
|---|---|
| Universe | Dow 30 constituents |
| Number of assets | 30 |
| Sample after lags | 2021-02-26 to 2026-01-26 |
| IV data source | AlphaQuery 30-day IV Mean |
| Neural model family | GNNHAR1L/2L/3L with and without IV |
| Training dates | 863 |
| Validation dates | 185 |
| Test dates | 186 |
| Graph construction | GLASSO on training returns |
| GLASSO alpha | 0.159968 |
| Graph edge density | 0.442222 |
| Neural base epochs | 250 |
| GNN hyperparameter selection | Validation split |
| MCS bootstrap replications | 60 |

# 4 Data and Experimental Design

The empirical analysis uses Dow 30 constituents with realized volatility, implied volatility, and daily returns. The implied-volatility input is sourced from AlphaQuery's volatility and option statistics pages (AlphaQuery, 2026). Specifically, the IV channel uses the 30-day "Implied Volatility (Mean)" field, which AlphaQuery reports as the average of put and call implied volatilities for options with the relevant expiration date. AlphaQuery also reports the corresponding 30-day call IV, put IV, put-call IV ratio, skew, and historical-volatility measures; this study uses the mean IV series as the primary forward-looking option-market signal. After HAR lag construction, the sample contains 1234 trading dates from 2021-02-26 to 2026-01-26. The chronological split uses 863 training dates, 185 validation dates, and 186 test dates. All reported tables are produced by the repository pipeline on branch `2026-06-01`. The tuned run uses 250 base neural training epochs, 60 MCS bootstrap replications, seed 20260602, validation-based GNN hyperparameter selection, and MSE-trained neural forecasts. The exported tables contain HAR, GHAR, GNNHAR1L, GNNHAR2L, GNNHAR3L, their IV extensions, IV placebos, random-graph controls, and the GNN hyperparameter-selection table.

# 5 Empirical Results

## 5.1 Main Forecast Comparison

Table 2 reports the main out-of-sample comparison. GHAR-IV is the best model by QLIKE among economically interpretable GLASSO specifications, while validation-tuned GNNHAR3L-IV is the best model by MSE. Relative to HAR, GHAR-IV reduces MSE by about 8.86% and QLIKE by about 4.51%. GNNHAR3L-IV reduces MSE by about 14.90% relative to HAR and by about 6.63% relative to HAR-IV, while its QLIKE is close to GHAR-IV. The placebo relation results require a more careful interpretation. The linear GHAR-IV-random model is nearly tied with GLASSO GHAR-IV, with QLIKE 0.0007843 versus 0.0007849. This means the linear GHAR-IV

Table 2: Out-of-sample forecast performance

| Model | Test MSE | Test QLIKE | MSE ratio/HAR | QLIKE ratio/HAR |
|---|---|---|---|---|
| GHAR-IV-random | 1.0272 | 0.0007843 | 0.9114 | 0.9541 |
| GHAR-IV | 1.0272 | 0.0007849 | 0.9114 | 0.9549 |
| HAR-IV | 1.0464 | 0.0007928 | 0.9285 | 0.9645 |
| GNNHAR3L-IV | 0.9591 | 0.0007964 | 0.8510 | 0.9689 |
| GHAR | 1.0848 | 0.0008111 | 0.9626 | 0.9868 |
| GNNHAR2L-IV | 1.0180 | 0.0008120 | 0.9032 | 0.9878 |
| GHAR-fakeIV | 1.0897 | 0.0008136 | 0.9669 | 0.9898 |
| GHAR-random | 1.0901 | 0.0008142 | 0.9672 | 0.9905 |
| HAR-fakeIV | 1.1217 | 0.0008207 | 0.9953 | 0.9985 |
| HAR | 1.1270 | 0.0008220 | 1.0000 | 1.0000 |
| GNNHAR3L-IV-random | 1.1733 | 0.0009459 | 1.0411 | 1.1508 |

Notes: The table reports test-period averages across dates and assets. The suffix "fakeIV" denotes a date-permuted IV control. The suffix "random" denotes a density-matched random graph placebo. GNNHAR rows use hyperparameters selected on the validation split, not on the test split.

Table 3: GNNHAR and GNNHAR-IV forecast performance

| Model | Estimation | Test MSE | Test QLIKE | QLIKE ratio/HAR |
|---|---|---|---|---|
| GNNHAR3L-IV | MSE | 0.9591 | 0.0007964 | 0.9689 |
| GNNHAR2L-IV | MSE | 1.0180 | 0.0008120 | 0.9878 |
| GNNHAR1L-IV | MSE | 1.0369 | 0.0008348 | 1.0156 |
| GNNHAR1L | MSE | 1.1485 | 0.0008476 | 1.0312 |
| GNNHAR2L | MSE | 1.1326 | 0.0008616 | 1.0483 |
| GNNHAR1L-IV-fakeIV | MSE | 1.2279 | 0.0008866 | 1.0786 |
| GNNHAR3L | MSE | 1.1905 | 0.0008962 | 1.0903 |
| GNNHAR3L-IV-random | MSE | 1.1733 | 0.0009459 | 1.1508 |

table alone cannot prove that the GLASSO relation matrix is economically identified. By contrast, GNNHAR3L-IV-random has QLIKE 0.0009459, much worse than GLASSO GNNHAR3L-IV, indicating that the nonlinear graph extension is much more dependent on the estimated relation matrix.

## 5.2 GNNHAR and GNNHAR-IV

Table 3 reports the validation-tuned neural variants. The key point is that GNNHAR-IV is explicitly estimated and tuned using the validation split: the table includes one-, two-, and three-layer GNNHAR models and their IV counterparts. Among the neural IV models, GNNHAR3L-IV has the lowest test MSE, 0.9591, and QLIKE 0.0007964. GNNHAR2L-IV is also competitive, with MSE 1.0180 and QLIKE 0.0008120. The IV channel is important: the corresponding no-IV neural models have higher losses. The random relation control is especially informative here. Replacing the GLASSO graph in GNNHAR3L-IV with a density-matched random graph raises MSE from 0.9591 to 1.1733 and QLIKE from 0.0007964 to 0.0009459. Thus, unlike the linear GHAR-IV case, the leading nonlinear GNNHAR-IV result is not reproduced by a randomized relation matrix.

Table 4: Model confidence set at the 5 percent level

| Model | Included | MCS p-value |
|---|---|---|
| GHAR-IV-random | Yes | 1.000 |
| HAR-IV | Yes | 0.833 |
| GHAR-IV | Yes | 0.833 |
| GNNHAR3L-IV | Yes | 0.833 |
| HAR | Yes | 0.467 |
| HAR-fakeIV | Yes | 0.400 |
| GNNHAR2L-IV | Yes | 0.400 |
| GHAR | Yes | 0.333 |
| GHAR-random | Yes | 0.250 |
| GHAR-fakeIV | Yes | 0.217 |
| GNNHAR1L-IV | Yes | 0.217 |
| GNNHAR3L-IV-random | No | 0.033 |

Table 5: Diebold-Mariano tests based on QLIKE losses

| Comparison | Mean loss A | Mean loss B | DM statistic / p-value |
|---|---|---|---|
| HAR vs GHAR | 0.000822 | 0.000811 | 0.467 / 0.641 |
| GHAR vs GNNHAR1L | 0.000811 | 0.000848 | -1.525 / 0.129 |
| GHAR-IV vs GNNHAR1L-IV | 0.000785 | 0.000835 | -2.175 / 0.031 |
| HAR-IV vs HAR-fakeIV | 0.000793 | 0.000821 | -1.837 / 0.068 |
| GNNHAR1L-IV vs GNNHAR2L-IV | 0.000835 | 0.000812 | 1.531 / 0.127 |

### 5.3 Model Confidence Set and DM Tests

Table 4 reports the MCS results at the 5% level. The confidence set contains GHAR-IV-random, GHAR-IV, HAR-IV, GNNHAR3L-IV, GNNHAR2L-IV, HAR, GHAR, and the main fake-IV controls. The high p-value of GHAR-IV-random is not interpreted as economic evidence for a random network. Instead, it shows that in the linear GHAR-IV model, the graph feature block can perform well even when the relation matrix is randomized. The nonlinear placebo behaves differently. GNNHAR3L-IV-random is excluded from the 5% MCS, while GLASSO GNNHAR3L-IV remains in the confidence set.

Table 5 reports targeted DM comparisons. HAR versus GHAR is not statistically significant at conventional levels. The comparison between HAR-IV and HAR-fakeIV has $p = 0.0679$, which is suggestive at the 10% level and supports the presence of real IV information. The comparisons involving tuned GNNHAR models show a more nuanced picture: GNNHAR1L is not significantly worse than GHAR at conventional levels, while GNNHAR1L-IV has higher QLIKE than GHAR-IV. GNNHAR2L-IV improves on GNNHAR1L-IV, but the difference is not statistically significant at the 5% level.

### 5.4 Does IV Contain Genuine Information?

The fake-IV decomposition in Table 6 is the central extension beyond Zhang et al. For HAR and GHAR, most of the total IV improvement comes from genuine information gain. For HAR, the total QLIKE improvement is $2.915 \times 10^{-5}$, of which $2.789 \times 10^{-5}$ is attributed to genuine

Table 6: IV information decomposition based on QLIKE

| Family | Baseline | Real IV | Fake IV | Total | Info | Param. |
|---|---|---|---|---|---|---|
| HAR | 0.0008220 | 0.0007928 | 0.0008207 | 0.0000292 | 0.0000279 | 0.0000013 |
| GHAR | 0.0008111 | 0.0007849 | 0.0008136 | 0.0000262 | 0.0000287 | -0.0000025 |
| GNNHAR1L | 0.0008476 | 0.0008348 | 0.0008702 | 0.0000128 | 0.0000354 | -0.0000226 |

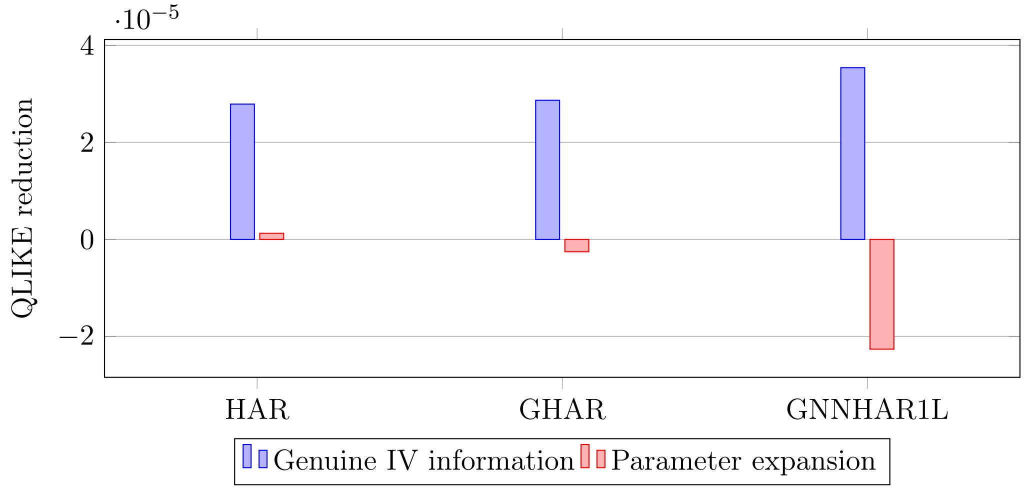


Figure 1: Decomposition of IV gains into genuine information and parameter expansion.

information. For GHAR, the genuine information gain is $2.868 \times 10^{-5}$. The parameter expansion term is small for HAR and negative for GHAR, meaning that fake IV does not explain the real-IV gains.

For GNNHAR1L, real IV also improves QLIKE relative to the no-IV neural baseline, and the fake-IV control is worse than real IV. This suggests that the IV channel contains information for the neural model as well, although the strongest neural result comes from the deeper GNNHAR3L-IV specification rather than the one-layer decomposition benchmark.

### 5.5 Regime-Specific Performance

Table 7 reports QLIKE losses in calm and volatile regimes. The volatile regime is defined as the top quartile of market-average realized volatility in the test period. GHAR-IV is the best GLASSO linear model in the calm regime, while GNNHAR3L-IV is the best model in the volatile top quartile. Its volatile-regime QLIKE is 0.001048, compared with 0.001168 for GHAR-IV, 0.001187 for HAR-IV, and 0.001244 for HAR. The random GNNHAR3L-IV placebo has volatile-regime QLIKE 0.001321, confirming that the stress-regime gain of the nonlinear model is tied to the GLASSO relation matrix rather than a generic random feature expansion.

### 5.6 Visual Summary

Figure 2 summarizes QLIKE ratios relative to HAR for the main linear models. Values below one indicate stronger out-of-sample QLIKE performance.

Table 7: Regime-specific QLIKE losses

| Model | Calm QLIKE | Volatile QLIKE |
|---|---|---|
| GHAR-IV | 0.000655 | 0.001168 |
| GHAR-IV-random | 0.000656 | 0.001163 |
| HAR-IV | 0.000660 | 0.001187 |
| GNNHAR3L-IV | 0.000711 | 0.001048 |
| GHAR | 0.000676 | 0.001211 |
| GHAR-random | 0.000678 | 0.001218 |
| GHAR-fakeIV | 0.000678 | 0.001215 |
| HAR-fakeIV | 0.000678 | 0.001241 |
| HAR | 0.000679 | 0.001244 |
| GNNHAR3L-IV-random | 0.000819 | 0.001321 |

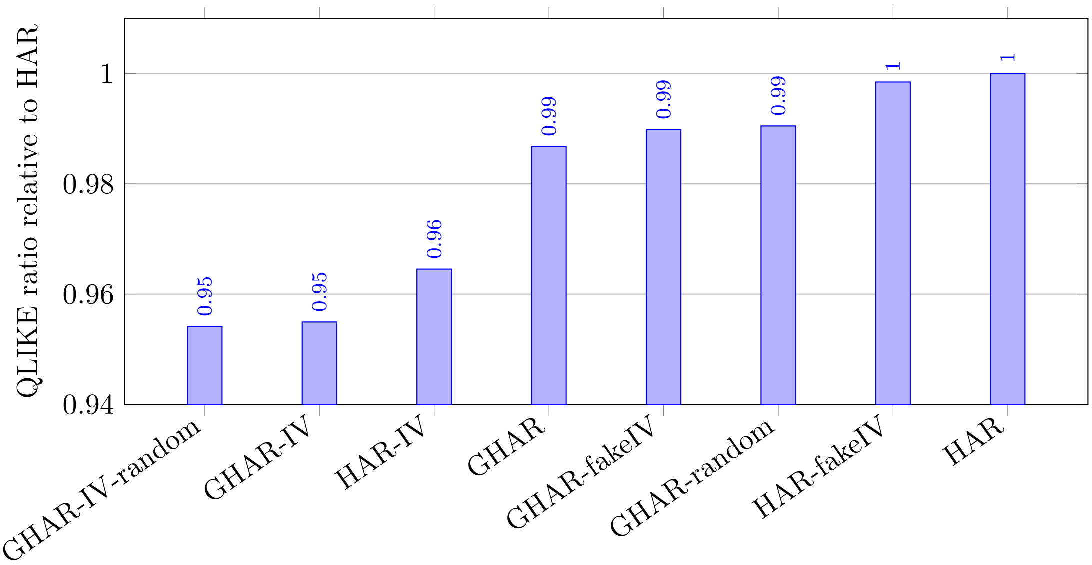


Figure 2: Out-of-sample QLIKE ratios relative to HAR for the main linear model family.

### 5.7 Distributional Diagnostics Following Zhang et al.

Zhang et al. use grouped boxplots of forecast errors and forecast ratios to diagnose whether models over-predict or under-predict realized volatility across calm and turbulent regimes. We replicate this diagnostic in Figure 3. The left column reports $\widehat{v}_{i,t} - v_{i,t}$, while the right column reports $\widehat{v}_{i,t}/v_{i,t}$. The dashed vertical lines mark zero forecast error and a unit forecast ratio. In the present Dow 30 experiment, the linear HAR/GHAR family remains tightly centered around these benchmarks, and the validation-tuned GNNHAR-IV variants remain competitive rather than collapsing into the much wider error distributions seen before tuning. This visual evidence is consistent with the loss tables: IV is valuable in the linear graph family and can also help nonlinear graph models when their hyperparameters are selected on the validation window.

Figures 4 and 5 report residual and calibration diagnostics. Figure 4 shows standardized residual Q-Q plots for HAR, HAR-IV, GHAR, and GHAR-IV. The four curves have similar central behavior and heavy right tails, indicating that IV and graph terms reduce average forecast loss without eliminating tail forecast errors. Figure 5 compares realized RV with forecast RV;

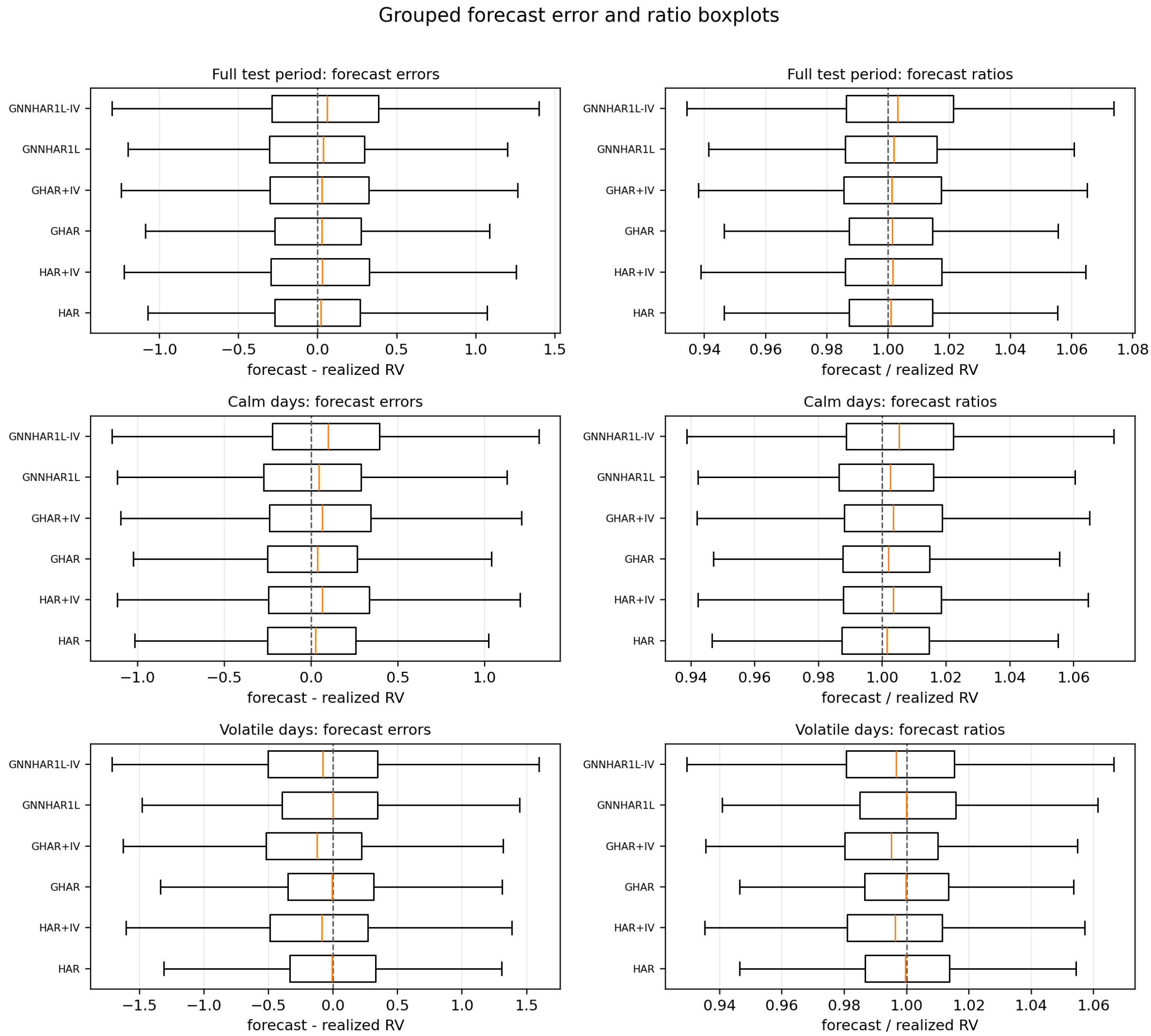


Figure 3: Grouped forecast error and forecast ratio boxplots following the diagnostic layout of Zhang et al. The rows report the full test period, calm days, and volatile days.

points close to the $45^\circ$ line correspond to well-calibrated forecasts.

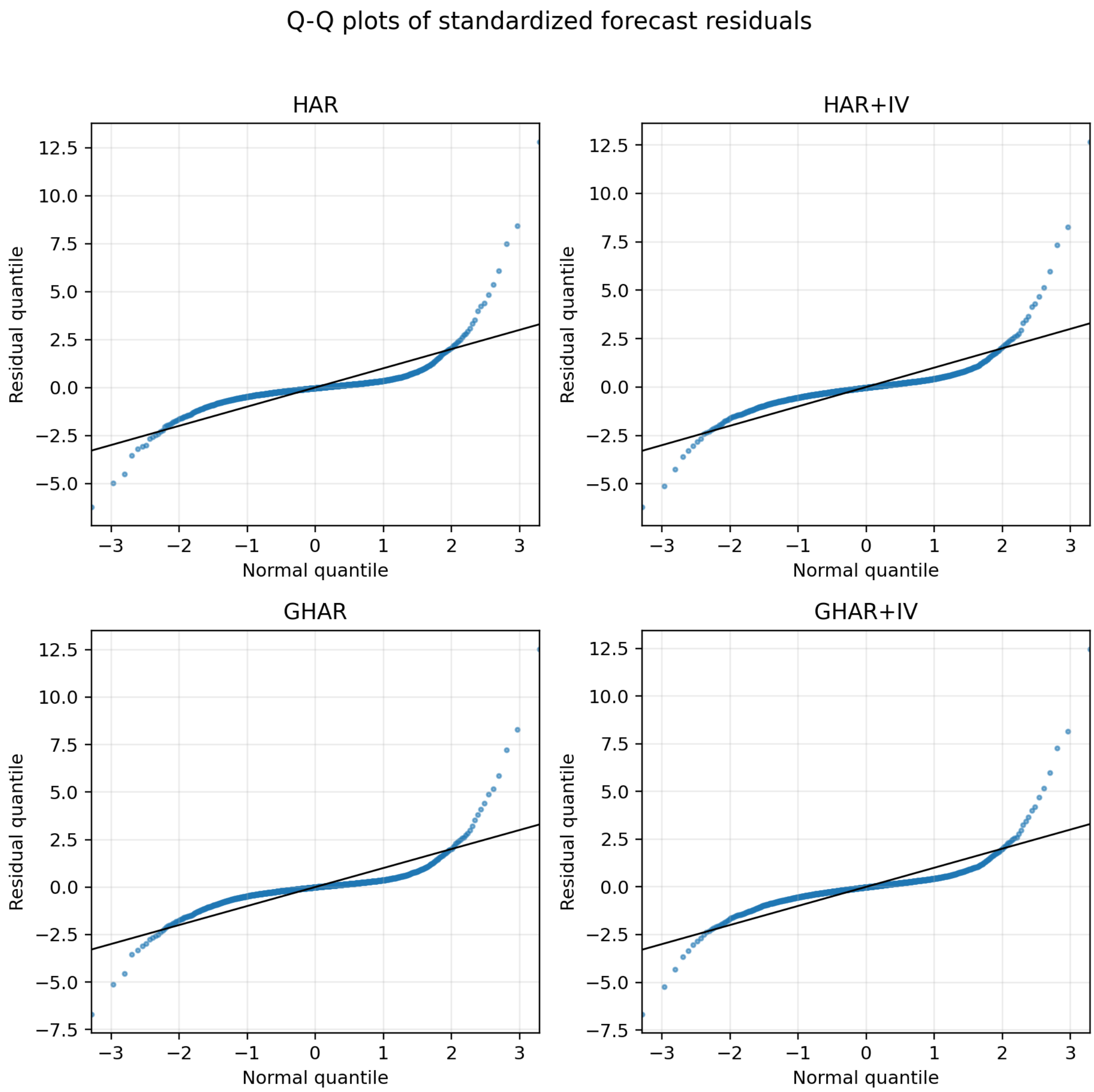


Figure 4: Q-Q plots of standardized forecast residuals for the main linear specifications.

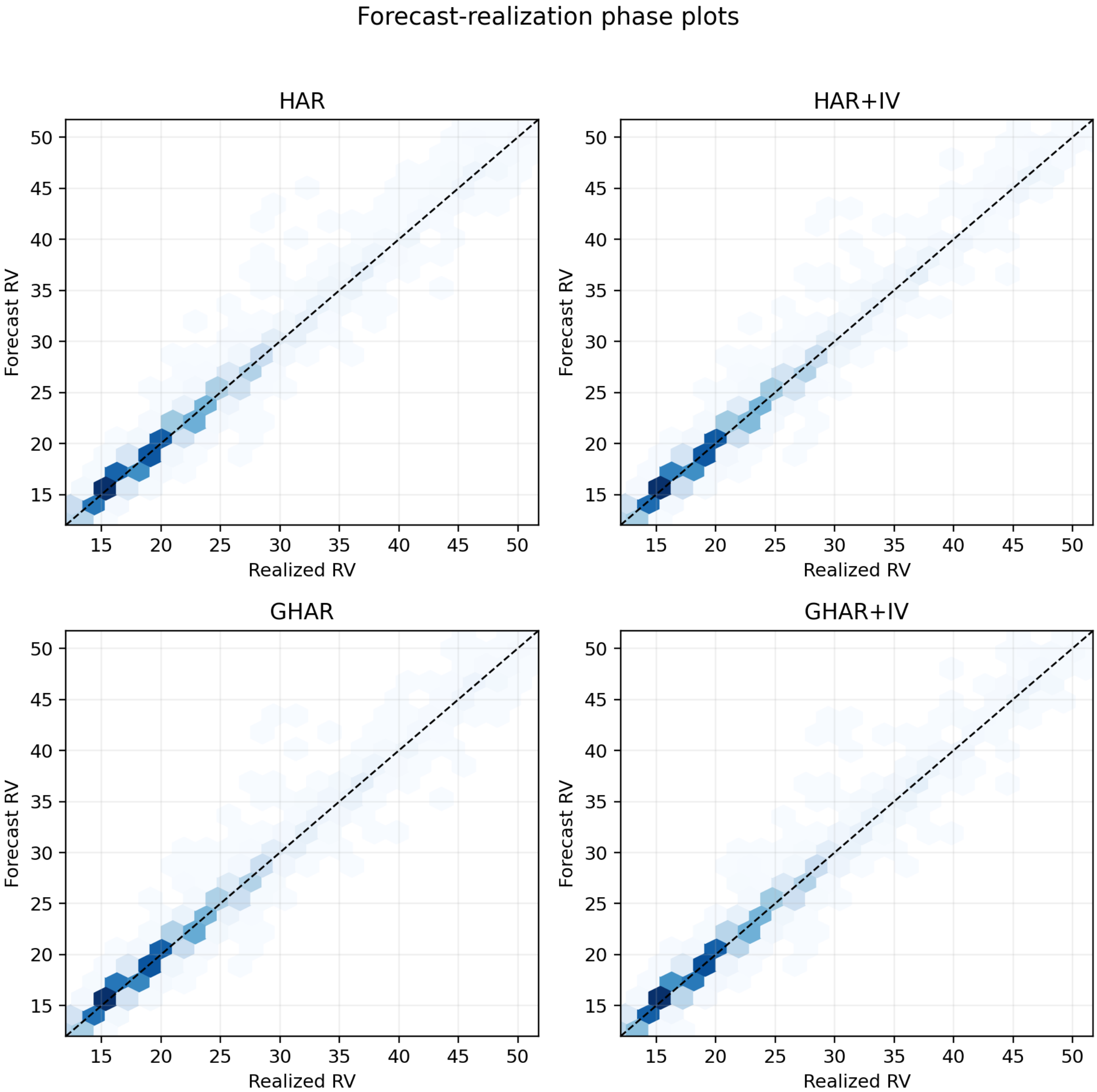


Figure 5: Forecast-realization phase plots for the main linear models.

# 6 Discussion

The empirical evidence supports three conclusions. First, implied volatility adds predictive content to graph-spillover realized-volatility models. The most direct evidence comes from the HAR and GHAR fake-IV comparisons, where real IV dominates fake IV. This matters because it rules out the simplest explanation that gains arise only because IV models contain more features.

Second, graph spillovers and IV are complementary in the linear family for prediction, but the relation interpretation is not fully identified by the linear table alone. GHAR improves on HAR, HAR-IV improves on HAR, and GHAR-IV improves on both. However, GHAR-IV-random is nearly tied with GLASSO GHAR-IV. This means that the linear graph block may partly work as a transformed-feature expansion. The appropriate conclusion is therefore not that the random graph is meaningful, but that linear GHAR-IV is not sufficient by itself to certify GLASSO relation information.

Third, nonlinear graph neural networks are sensitive to tuning and to the relation matrix in the Dow 30 setting. Zhang et al. study a larger universe and report that nonlinear GNNHAR models can improve forecasting accuracy. In the present Dow 30 run, validation-tuned GNNHAR3L-IV delivers the lowest MSE and the best volatile-regime QLIKE, while GHAR-IV remains the best economically interpretable GLASSO linear model by full-sample QLIKE. Crucially, GNNHAR3L-IV-random is much worse than GLASSO GNNHAR3L-IV. This is the main evidence in the paper that the estimated GLASSO relation matrix matters beyond generic graph-feature expansion.

# 7 Robustness and Limitations

Several robustness checks are built into the analysis. First, fake-IV controls show that real-IV gains are not explained by feature dimension alone in HAR and GHAR. Second, random-graph controls separate relation information from transformed-feature expansion. These controls give mixed but informative evidence: GHAR-IV-random is nearly tied with GLASSO GHAR-IV, whereas GNNHAR3L-IV-random is substantially worse than GLASSO GNNHAR3L-IV and is excluded from the 5% MCS. Third, the regime split shows that GHAR-IV remains strong in calm periods, while GNNHAR3L-IV performs best in the volatile top quartile. Fourth, the GNN tuning table records validation MSE, validation QLIKE, and the selected hidden size, learning rate, and epoch budget for each neural specification.

The study has limitations. The Dow 30 universe is much smaller than the S&P 100 setting in Zhang et al., so the graph neural networks have less cross-sectional information. The IV data are taken from AlphaQuery's summarized option-statistics fields rather than reconstructed from raw option chains; therefore, vendor definitions, data availability, and interpolation choices may affect results. The neural models use a compact validation grid rather than a very large hyperparameter sweep. Finally, neural results can remain subject to runtime stochasticity even with fixed seeds; the reported tuned tables are therefore tied to the stated seed, split, and hyperparameter-selection protocol.

## 8 Conclusion

This paper develops and evaluates a GNNHAR-IV extension of Zhang et al.'s graph-spillover realized-volatility forecasting framework. The extension adds implied volatility at the same HAR horizons as realized volatility and evaluates whether the IV channel contributes genuine information beyond parameter expansion. The empirical results are strongest for IV: real IV dominates fake IV in HAR and GHAR. The graph evidence is deliberately more qualified. Linear GHAR-IV forecasts are strong, but the random-relation placebo is nearly tied with the GLASSO relation, so the linear relation channel should not be over-interpreted. The nonlinear extension gives sharper relation evidence: GNNHAR3L-IV achieves the lowest test MSE and the best volatile-regime QLIKE, while its random-graph counterpart performs poorly. Overall, IV is a useful addition to the GNNHAR framework, and IV-placebo plus relation-placebo validation is necessary for interpreting the source of the forecasting gains.

## A Reproducibility Appendix

The empirical results are generated from the project repository rather than from hand-edited tables. The repository contains the notebook `notebooks/gnnhar_iv_colab_full_run.ipynb`, which runs the same pipeline used for the reported tables and figures. The notebook can be opened in Colab through:
Colab notebook for the GNNHAR-IV replication run.

For local replication, the main command is:

```
python scripts/analysis/gnnhar_iv_pipeline.py -data-dir experiments/dow30/data
-output-dir outputs/gnnhar_iv_relation_tuned_full -epochs 250 -tune-gnn
-mcs-bootstrap 60 -skip-qlike-training.
```

The run uses seed 20260602, 1234 usable dates, and the 863/185/186 train-validation-test split reported in Table 1. The key output files are:

- `tables/model_losses.csv`: out-of-sample MSE, QLIKE, and relative loss ratios.
- `tables/gnn_tuning.csv`: validation-selected hidden size, learning rate, and epoch budget for each GNNHAR specification.
- `tables/iv_decomposition.csv`: fake-IV decomposition into information and parameter-expansion components.
- `tables/mcs_results.csv` and `tables/dm_tests.csv`: MCS and pairwise forecast-loss tests.
- `figures/`: boxplots, Q-Q plots, phase plots, IV decomposition, and graph diagnostics.

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